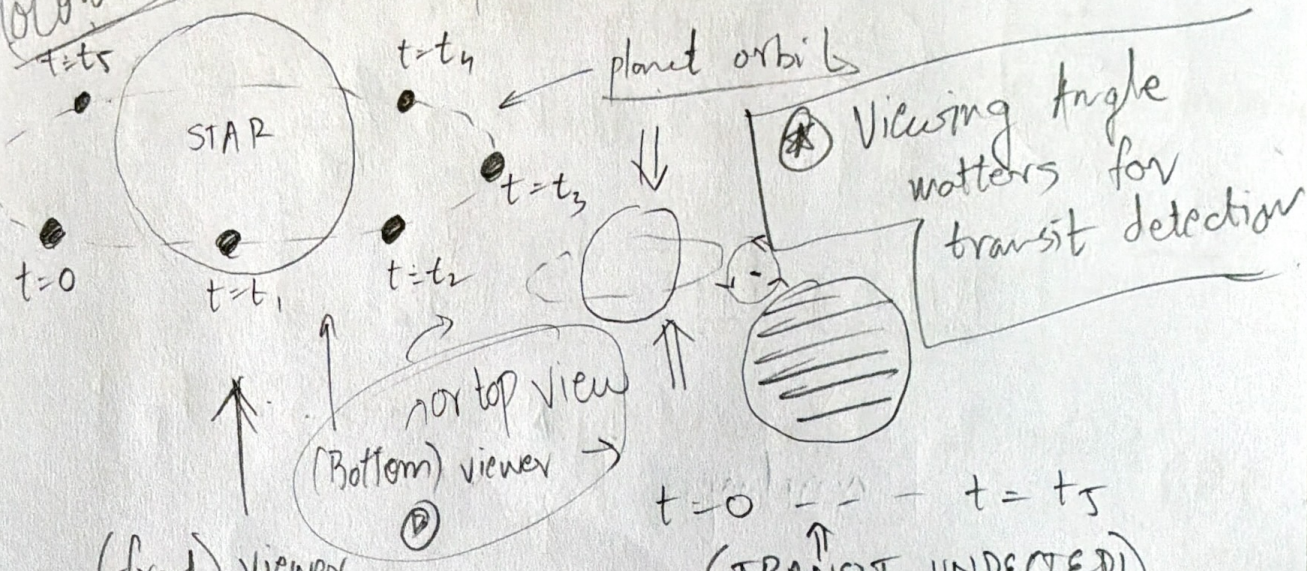
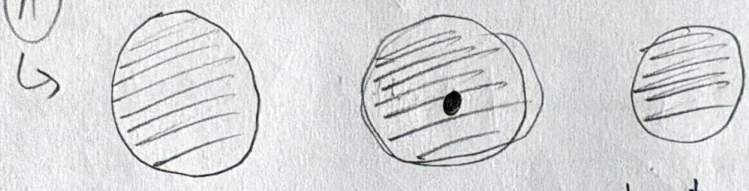
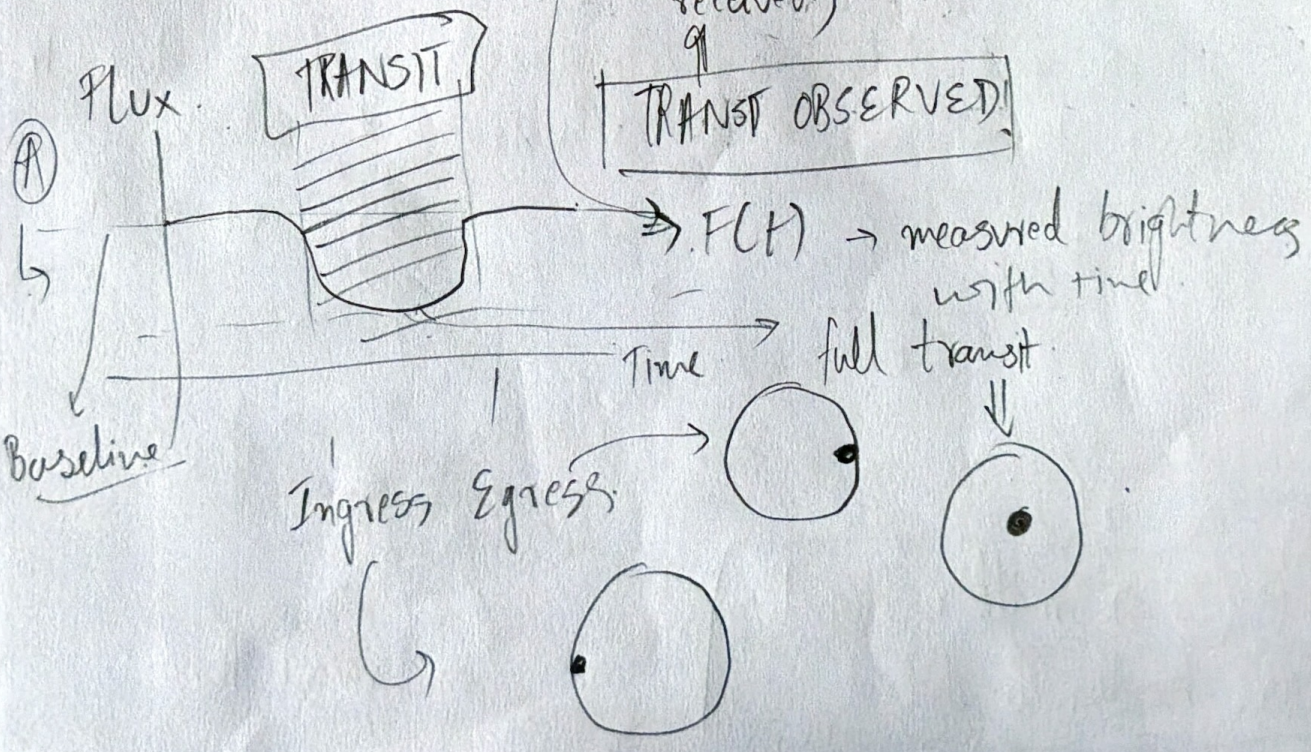
TESS TRANSIT DETACTABILITY.

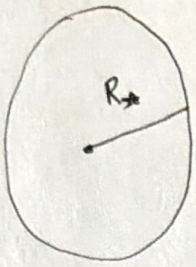
Molonetry

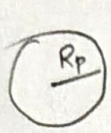






(Projected) $A_\star = \pi R_\star^2$





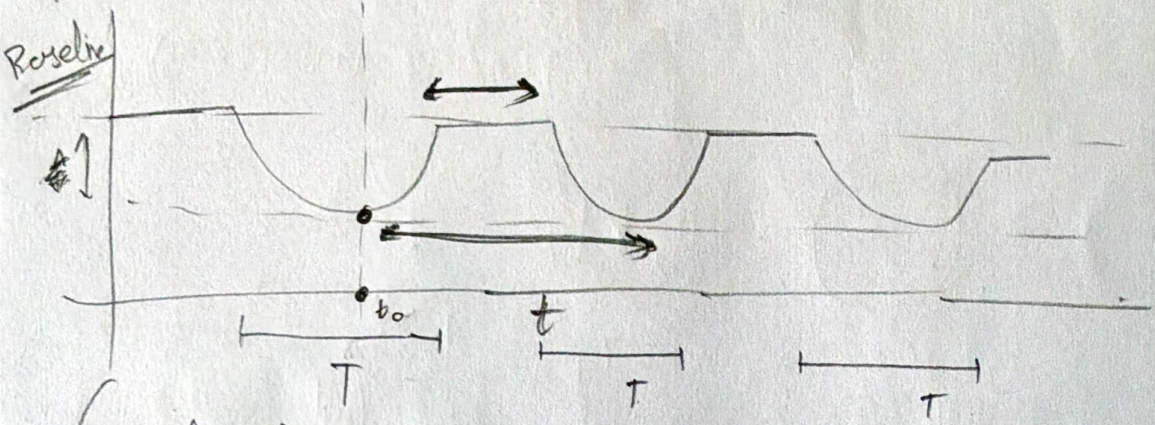
(Project) $A_p = \pi (R_p)^2$

Planet: Light blocked $\Rightarrow \frac{A_p}{A_\star} = \delta$

Star

TRANSIT DEPTH $\leftarrow \quad \delta \approx \left(\frac{R_p}{R_\star}\right)^2$ ——— ①

$F(t)$ > Given $R_\star$, δ (measured by Δbrightness) $\rightarrow R_p$!

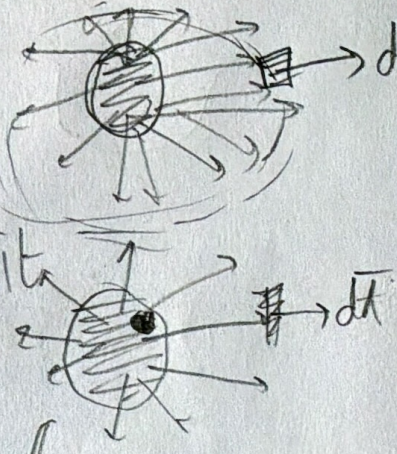


only for Star – Planet system

⇓

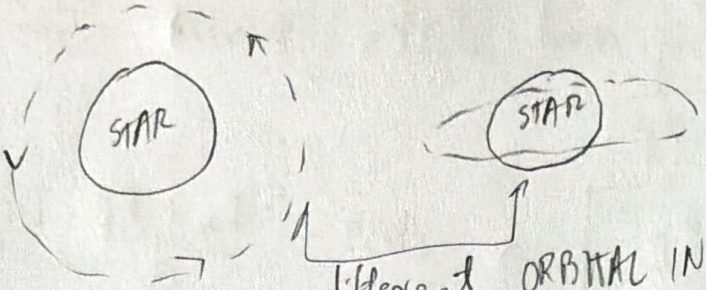
in real data → δ (Δ Brightness) could be due to other stars and matter instrument failure.

$$\delta = \frac{F_{out} - F_{in}}{F_{out}}$$ → ②



① ② ⇒ Smaller Planets are harder to detect.

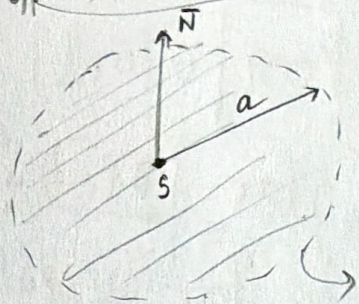
3.



different ORBITAL INCLINATION (i) w.r.t observe
↳ effects the detectability.

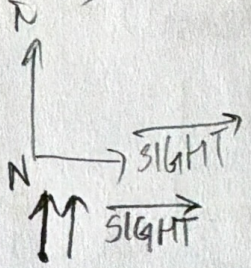
---→ ORBIT

(CIRCULAR ORBITAL GEOMETRY) →



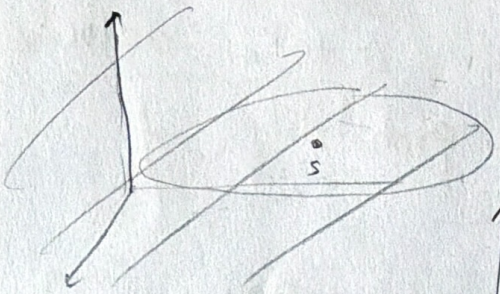
Edge-on i = 90°

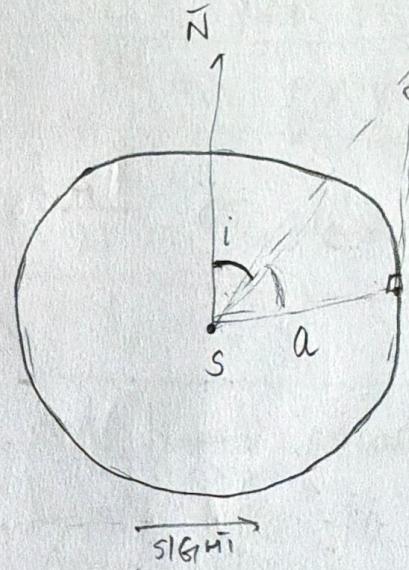
Face-on i = 0

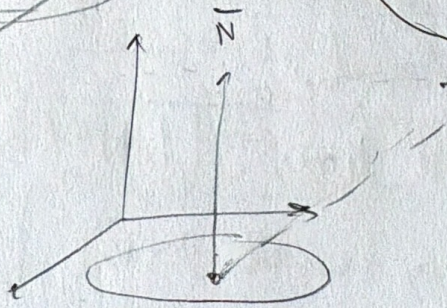


Orbital Plane
sight

for general i:







$$d_{proj} = a\cos i$$

impact parameter ⇒ $b = \frac{d_{proj}}{R_\star} \Rightarrow \frac{a\cos i}{R_\star}$

↳ how far the planet's centre passes from stellar disk

$b = 0$ → Planet crosses centre of star

$b \to 1$: closer to crossing near star edge.

(11/2) for transit, projected disks must overlap:

$a\cos i < R_\star + R_p$

$$|b| < 1 + \frac{R_p}{R_\star} \rightarrow \cos i < \frac{R_\star + R_p}{a}$$

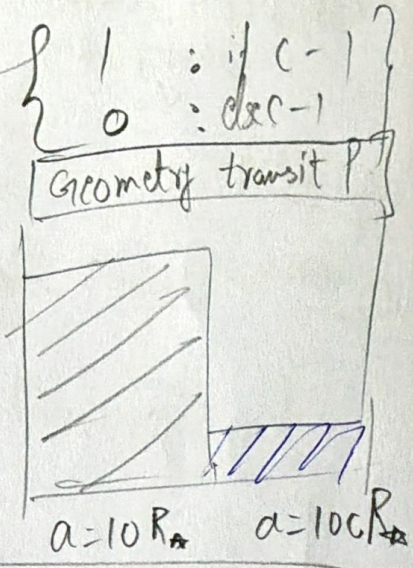
$$|\cos i| \in [0,1] \Rightarrow \cos i \sim \mathcal{U}(0,1)$$

or for PDF: $P(\cos i) = \begin{cases} 1 & : \text{if } c < 1 \\ 0 & : \text{else} \end{cases}$

$c = 1$; for $0 \leq \cos i \leq 1$

for very small planet,

$$P_{transit} \propto \frac{R_\star}{a}.$$



(X) Far away planet has smaller probability of lining up with planet.

We can model transit times: LINEAR TRANSIT EPHEMERIS

$t_n = t_0 + nP$ → transit period

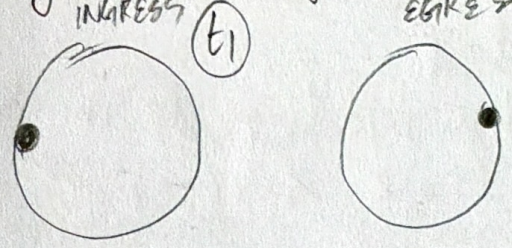
↳ reference transit midpoint

(5) $g = \frac{G M_\star M_p}{a^2} \rightarrow F_c = \frac{M_p v^2}{a} \Rightarrow v^2 = \frac{G M_\star}{a} \left(= \frac{2\pi a}{P}\right)$

we get: $P^2 = \frac{4\pi^2 a^3}{G M_\star}$ or $\underline{P^2 \propto a^3}$!

(6) More distant planets have greater orbital period

Assuming aided edge on orbit:



$L \Rightarrow 2[R_\star + R_p]$ ③ ↳ total dist. traversed

$v = \frac{2\pi a}{P}$ -- ⑦ ⟶ ③, ④ $\Rightarrow T_{14} \approx \frac{L}{v}$,

$T_{14} = \frac{2(R_\star + R_p)}{\frac{2\pi a}{P}}$

$\Rightarrow \boxed{T_{14} \approx \frac{P(R_\star + R_p)}{\pi d}}$

for $R_p << R_\star$ ⟶ $T_{14} = \frac{P R_\star}{\pi a}$

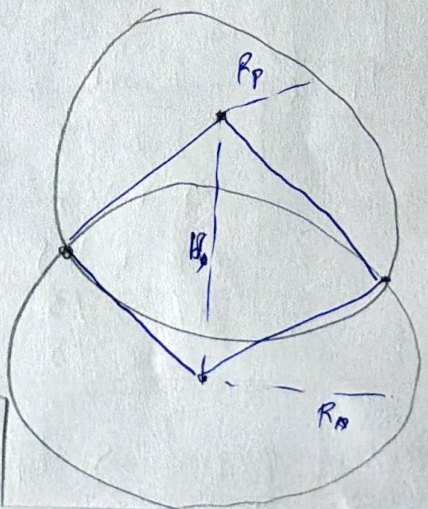
CENTRAL ORBITS ↑

for non-central orbits:

$d_{proj} = b R_\star$

$x^2 + b R_\star^2 = (R_\star + R_p)^2$

$\boxed{T_{14} = \frac{P}{\pi a}\left(\sqrt{(R_\star + R_p)^2 - b^2 R_\star^2}\right)}$



(5) larger $b \Rightarrow$ shorter transit

BLS: (Box least squares).

$N \leftrightarrow$ observations

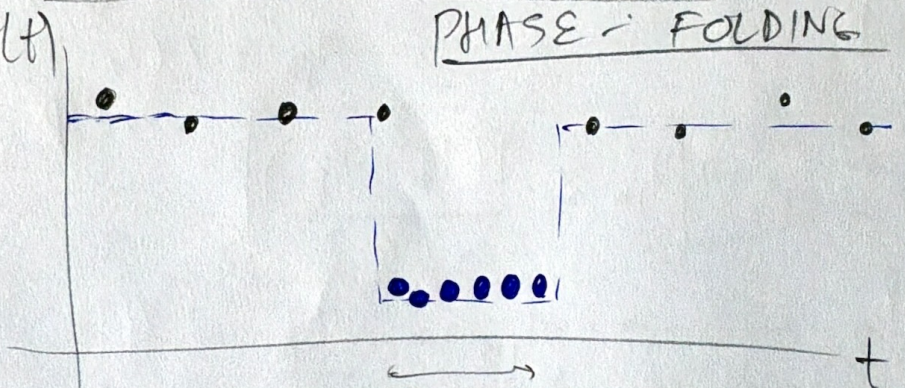
$$D = \{(t_i, F_i, \sigma_i)\}_{i=1}$$

t_i: obs time; F_i: stellar brightness; σ_i: uncertainty

Q: Is there a periodically repeating dip in brightness in such a noisy time series?

SIMPLIFIED MODEL PARAMETERS: $\theta = (P, t_0, T, \delta)$

PHASE - FOLDING



repeated transits overlap at the same location of t-axis

$$\phi_i = \left\{ \frac{t_i - t_0}{P} \right\}$$

suppose: $t = (2, 7, 12, 17)$ days

$\hookrightarrow$ suspicion $P = 5$ days

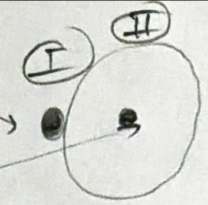
example	Observ. Time	work	Phase
Day	2	(2-2)/5	0
Day	7	(7-2)/5	0
	12	(12-2)/5	0
	17	(17-2)/5	0
	8	(8-2)/5	0.2

All 4 transits have phase zero.

* for incorrect period $\rightarrow$ dips stop lining up

① SIMPLE MODELING:

2 brightness levels: H: out of transit
L: in-transit



$\delta = H - L$ ✓ → indicator fcn: z_i = 1 : for i ∈ (transit), 0 : for i ∉ (transit)

Model $m_i = H - \delta z_i$ for repeating transit signal.

$z_i = 0 \rightarrow m_i = H$ ✓

$z_i = 1 \rightarrow H - \delta$ ✓

defined by param: P, T, t_0

Brightness → H, δ ✓

Predicted: m_i ; Measure → F_i

$e_i = F_i - m_i$ → goal: minimise e_i

Minimise!

$$S = \sum_{i=1}^{N} (F_i - m_i)^2$$

But better! Measurement uncertainties differ, so we minimise weighted:

$$\chi^2 = \sum_{i=1}^{N} \frac{(F_i - m_i)^2}{\sigma_i^2}$$

using m_i relation:

$$\chi^2 = \sum_{i=1}^{N} \frac{(F_i - H + \delta \cdot z_i)}{\sigma_i^2}$$

P, T, transit midpoint (t₀)

↓

we determine H, δ producing BEST FIT.

① Simple case ⇒ All uncertainties are same. i.e all measurements have equal uncertainty.

ASSUMPTION FOR NOW.

$\hat{H} = \overline{F_{out}}$ and $\hat{L} = \overline{F_{in}} \Rightarrow \hat{S} = \overline{F_{out}} - \overline{F_{in}}$

kill assumption.

↳ Now for unequal weights? → $w_i = \frac{1}{\sigma_i^2}$

$F_{in} = \frac{\sum_{(in)} w_i F_i}{\sum_{(in)} w_i}$

$F_{out} = \ldots$

$J = \ldots$

SAME STRUCTURE STILL!

CHI-SQUARE IMPROVEMENT

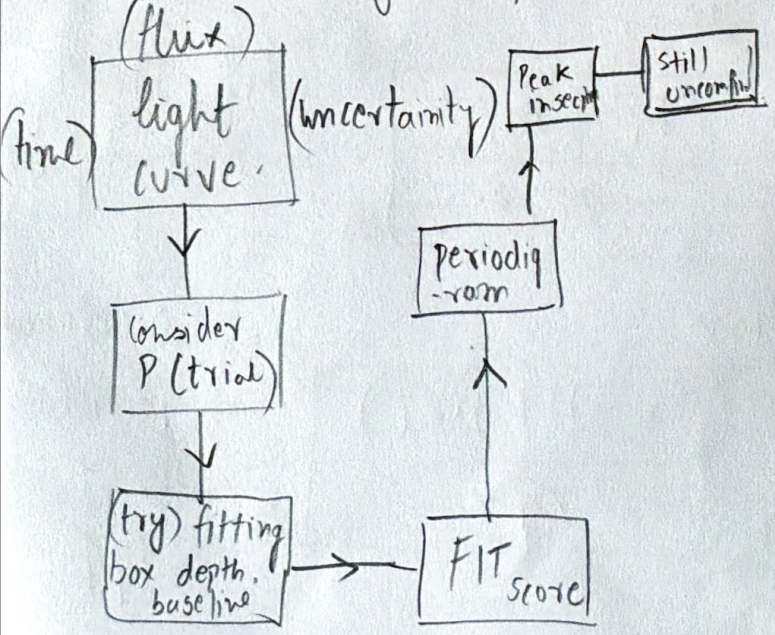
Model
- 1: Const. Brightness, X TRANSIT
- 2: Best fitting periodic box

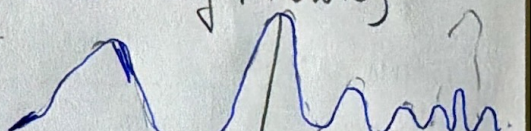
$\Delta\chi^2 = \chi^2_{flat} - \chi^2_{box}$

More +ve → box > flat!

(Good fit always planet? Maybe)

BLS searching for period. (P, T, t_0 × correctly known)





MORE PROMINENT PEAKS → MORE PROMISING PERIODICITIES.

9. Incorrect period interferes signals?

↳ T: 1, 6, 11 → corr(P) = 5

↳ phases are 0:

incorr(P) = 4

1	0
6	0.25
11	0.50

dips don't align anymore

* (Later) correction: some incorrect periods can produce convincing alignments too.

Objective: $\sum_i (F_i - H + \delta_{2i})^2$

let → $l = H - \delta$

$$S = \sum_{out} (F_i - H)^2 + \sum_{in} (F_i - l)^2$$

$$\frac{\partial S}{\partial H} = -2 \sum_{out} (F_i - H) = 0$$

↳ $\hat{H} = \bar{F}_{out}$

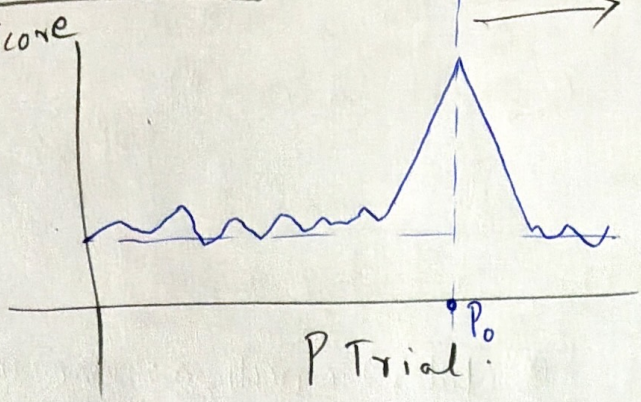
$$\frac{\partial S}{\partial H} = -2 \sum_{in} (F_i - l)$$

↳ $\hat{l} = \bar{F}_{in}$

$$\hat{\delta} = \bar{F}_{out} - \bar{F}_{in}$$

BLS PERIODGRAM

(10)



→ STRONG FIT

⇓

But we don't know correctly still if P_0 is actual period

SNR: $\delta = \frac{0.005}{\sigma}$ (e.g.) → star dims by 0.5%

↳ This observation has noise, assume std. dev (σ) of each noise ⇒ $\sigma = 0.01$

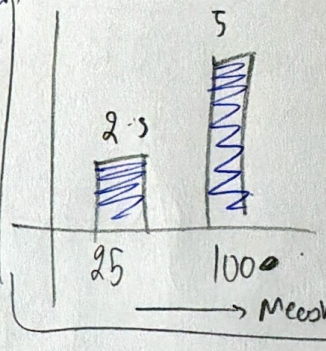
SNR ⇒ 0.5 → Very BAD

later add^{n}: don't rely only on 1 signal

$$Var(\bar{F}_{in}) = \frac{\sigma^2}{F_{in}} \rightarrow \sigma_{mean} = \frac{\sigma}{\sqrt{N_{in}}}$$

Clarification: only when independence is assumed

$$SNR \approx \frac{\delta \sqrt{N_{in}}}{\sigma}$$



4x independent measurement ⇒ 2x SNR

What happen if we include uncertainity in baseline? → $Var(\bar{F}_{out}) \Rightarrow \frac{\sigma^2}{N_{out}}$

$$\hat{\delta} = \bar{F}_{out} - F_{in}, \quad Var(\hat{\delta}) = \left(\frac{1}{N_{out}} + \frac{1}{N_{in}}\right)\sigma^2$$

$$SNR_{\delta} = \frac{\hat{\delta}}{\sigma\sqrt{\frac{1}{N_{out}} + \frac{1}{N_{in}}}}$$

→ getting closer to reality, baseline may not be 100% perfectly known.

(11) $W_{in} = \sum_{in} \frac{1}{\sigma_i^2}$, $W_{out} = \sum_{out} \frac{1}{\sigma_i^2}$ [Unequal importance uncertainty.]

Then ,

$$SNR_f = \frac{\hat{f}}{\sqrt{\frac{1}{W_{in}} + \frac{1}{W_{out}}}}$$ ✓

$\Delta\chi^2 = \frac{\hat{f}^2}{\sigma_f^2} \Rightarrow \Delta\chi^2 = \left(\frac{\hat{f}}{\sigma_f}\right)^2$?

$\Rightarrow \Delta\chi^2 = SNR_f^2$

IDEAL WHITE NOISE MODEL : → (Measurement errors at distinct times → uncorrelated)

$$Cov(\epsilon_i, \epsilon_j) = 0 \quad (i \neq j)$$

RED NOISE → (observations are correlated over time)

$Cov(\epsilon_i, \epsilon_j) \neq 0$ for distinct obs. times of course

↳ • STELLAR VARIABILITY

Candidate P	Rel ↔ true $P = P_t$	Interpretation
days: 6	P_t	
days: 3	$P_t/2$	integer rel. harmonic card
days: 12	$2P_t$	integer rel subharmonic card
days: P	Complex rel	Noise maybe

Maybe not planet

↳ Why BLS incorrect at times → eclipsing binary stars, stellar var., CONTAM.

HIGHEST PEAK NOT THE ONLY ANALYSIS

(12)

SYNTHETIC INJECTION RECOVERY OUTCOMES:

(inject transits, then try recovering them).

VI- cut.

DIRECT RECOVERY	: 98
HARMONIC RECOVERY	: 25
ALIAS RECOVERY	: 12
Non- Recovery	: 185

} 320

Q. IF WE INSERT A SIGNAL WHOSE TRUE PARAMETERS ARE KNOWN, can our detection procedure recover ~~them~~ it correctly?

---

for avg. N errors

$$Var(\bar{\varepsilon}) = \frac{1}{N^2}\left[\sum_i Var(\varepsilon_i) + 2\sum_{i<j} Cov(\varepsilon_i, \varepsilon_j)\right]$$

for independent noise, (Cov) $\times$ → we get $\frac{\sigma^2}{N}$ ↓↓.

But if errors are +vely correlated → Cov ↑↑ Variance.

(∴ CORRELATED NOISE BREAKS THE $\sqrt{T}$ Rule)

Synthetic Transit Injection and Recovery:

$\theta_{true} = (P, t_0, T, \delta)$ (INPUT) →→ [Detection] (without θ_{true}) → [output]

COMPARE ε . .

(3) SIMPLE INJECTION MODEL:

Considering ideal normalized stellar light curve:

$$F_0(t) \approx 1 + \varepsilon(t)$$

↳ noise, other variation.

let a synthetic box transit:

$$M(t) = \begin{cases} 1-\delta & : \text{in} \\ 1 & : \text{out} \end{cases}$$

$$F_{inj}(t) = F_0(t)\, M(t)$$

FAILURE #1:

The baseline must be represented correctly.

- Light curve was represented approx as: $f_{detrended}(t) \approx 0$

$f_{inj} = f_{detrended} M$ WRONG!

CORRECTIONS:

$$\delta_{raw} = 1 - \min(M_{raw})$$

$$\delta_{req.} \Rightarrow \frac{(depth)_{ppm}}{10^6}$$

$$M_{norm} = 1 - \frac{\delta_{req}}{\delta_{raw}}(1 - M_{raw})$$

FAILURE #2

- Generated TRANSIT MODELS with requested depth but achieved depth did not...

So, THE STATEMENT:
320 reject inj. = 320 failed detections.
WRONG!

$$f_{inj} = (1 + f_{detrended})\, M_{norm} - 1$$

$M_{norm,min} = 1 - \delta_{req}$

(14)

CHOOSING EXPERIMENTAL PARAMETERS

- injection grouped into 4 ranges:

P_1	0.3 – 1 day	orbital period
P_2	1 – 3 day	
P_3	3 – 10 day	
P_4	10 – 30 day	

Why?
↳ deep + transit → easy
shallow transit → difficult

Transit depth level: $2^n (10)^2$ → 100, 200, 400, 1600, 3200, 6400, 12800

Q. How does recovery ↔ depth ?

Detectability Matrix:

	0.3 – 1	1 – 3	3 – 10	10 – 30
100 ppm				
200 ppm				
400 ppm				
1600 ppm				
3200 ppm				
6400 ppm				
12800 ppm				

No. of combination ⇒ 32 cells

$$\hat{C}_{jk} = \frac{N_{direct\ jk}}{N_{valid,jk}}$$

(P) But what about uncertainty? include:

$n = N_{valid\ yk}$

let X ⇒ tot successful recoveries

$X \sim Bin(n, p)$

$E[X] = np$, $Var[X] = np(1-p)$

$\hat{c} = \frac{X}{n}$, $Var(\hat{c}) = Var\left(\frac{X}{n}\right) = \frac{1}{n^2} Var(X)$

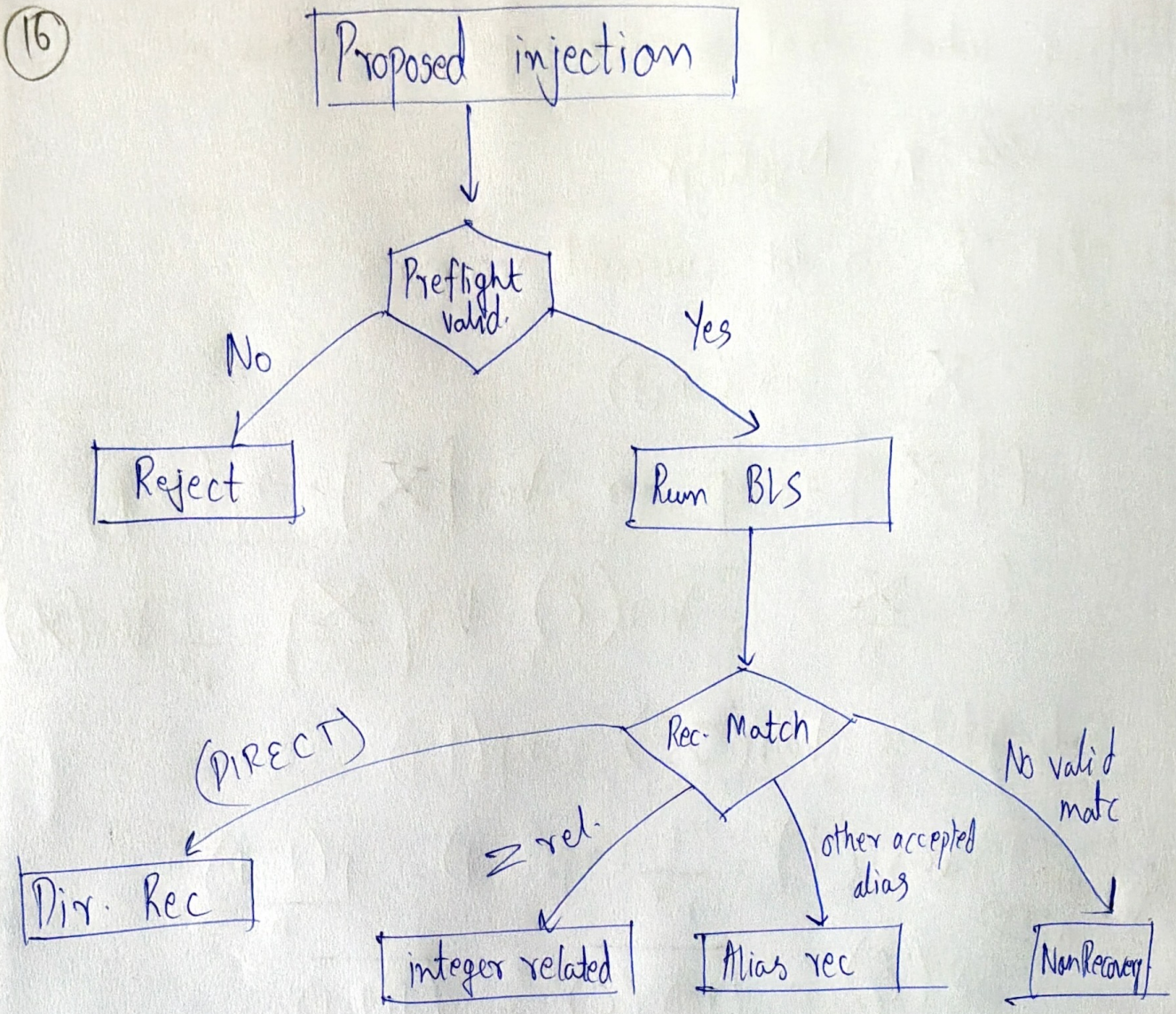
Substituting $Var(X) = np(1-p)$

Standard error ↑

$Var(\hat{c}) = \frac{1}{n^2} \cdot np(1-p) = \frac{p(1-p)}{n}$

$SE(\hat{c}) = \sqrt{Var(\hat{c})} = \sqrt{\frac{p(1-p)}{n}}$

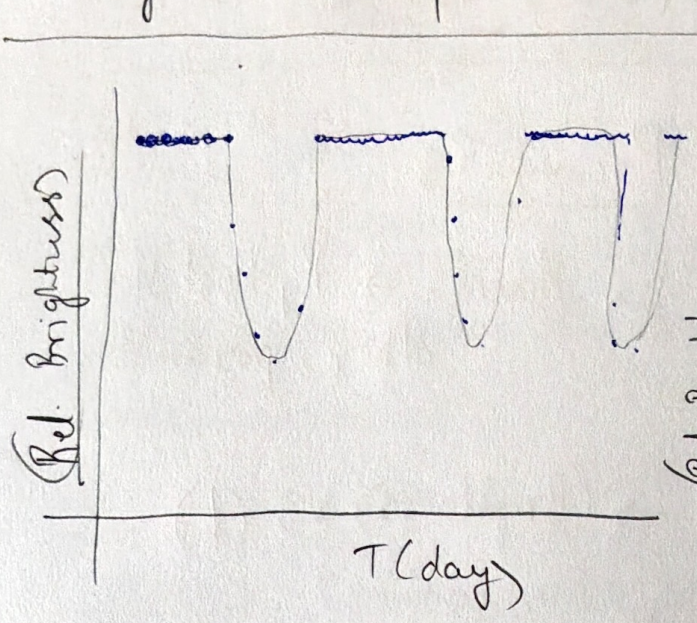
$\Rightarrow \sqrt{\frac{\hat{c}(1-\hat{c})}{n}}$



$$\varepsilon_p = \frac{|P_{rec} - P_{inj}|}{P_{inj}}$$

$$\Delta t = \min_{n \in \mathbb{Z}} \left| t_{rec} - (t_{inj.} + n P_{inj}) \right|$$

2 ways to test planet detection algorithm

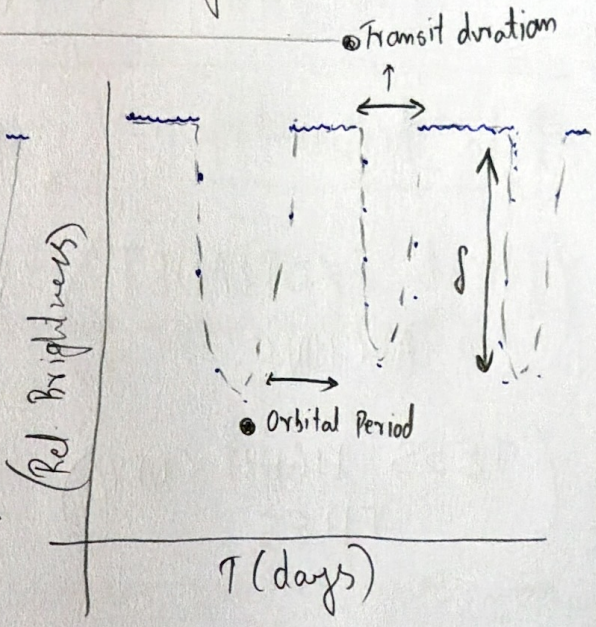


(A)

Real light curve containing previously catalogued planetry system.

Q: Can our algorithm find the known periodic signal?

KNOWN PLANET RECOVERY



(B)

Insert a controlled synthetic transit into an observed light curve

Q: Can our algorithm recover signal we added?

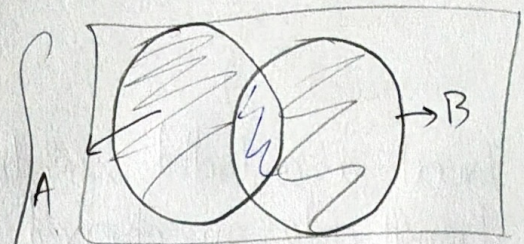
INJECTION RECOVERY

(6) FINDING A PERIOD IS NOT ENOUGH

Data Acquisition → Pscompors

(NASA EXOPLANET ARCHIVE) → planets ⇒ P, host stars, other parameters

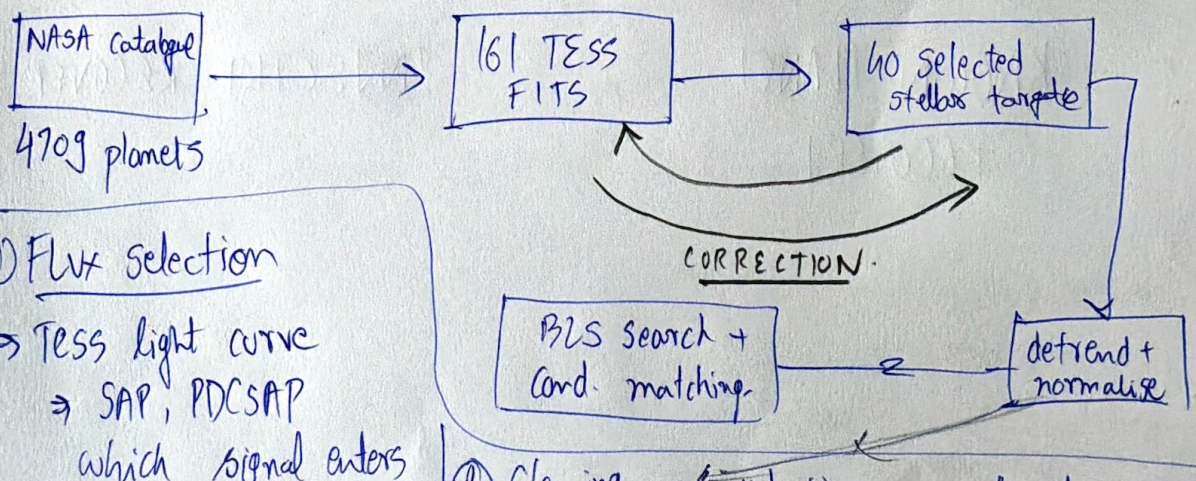
(TESS LIGHT CURVE FILES) → (Brightness) vs (t)



$N_{kept} = |A \cap B| \neq (A^c| - (B^c|$

A → Measurements failing with quality condition
B → Measurements ∉ with invalid t values

DATA PROCESSING PIPELINE:



① Flux selection
⇒ Tess light curve
⇒ SAP, PDCSAP
which signal enters analysis?

② Cleaning: invalid times or ∉ values, quality flags.

③ Detrending : Stellar brightness change not because of planetary transits

LEARNT : Too much detrending can erase a real transit.

~~Combine every sector into on long curve and apply the same correction.~~

↳ WRONG

CORRECTED : Reconsider sector boundaries, baselines, gaps, and observing conditions before combining observation

V.1.0.0 ⇒ $f_{inj} = f_{det} M$ (✗)

V.1.0.1 ⇒ $f_{inj} = (1 + f_{det}) M_{raw} - 1$ (✗)

V.1.0.2 ⇒ $M_{norm} = 1 - \frac{f_{req}}{f_{raw}} (1 - M_{raw})$ (✓)

Q. Can time correlated stellar noise help predict transit recovery beyond expected SNR and observing window info?

⇓

V2! →

- (✓) Question, comparison
- (✓) Data acquisation
- (✗) Large scale qualified exp (in complete)
- (✗) No V2 improvement